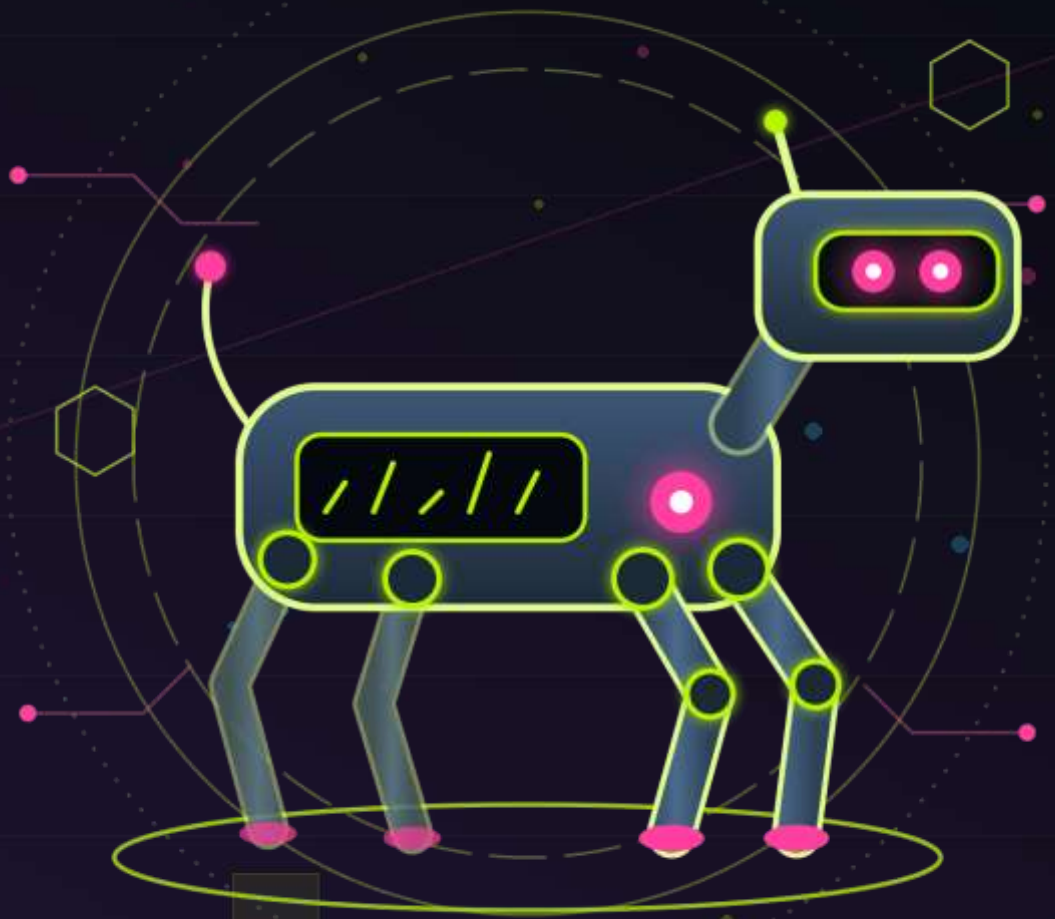
Python Seaborn for AI & Robotics



Statistical charts for robot sensor data: relationships, distributions, categories and grids

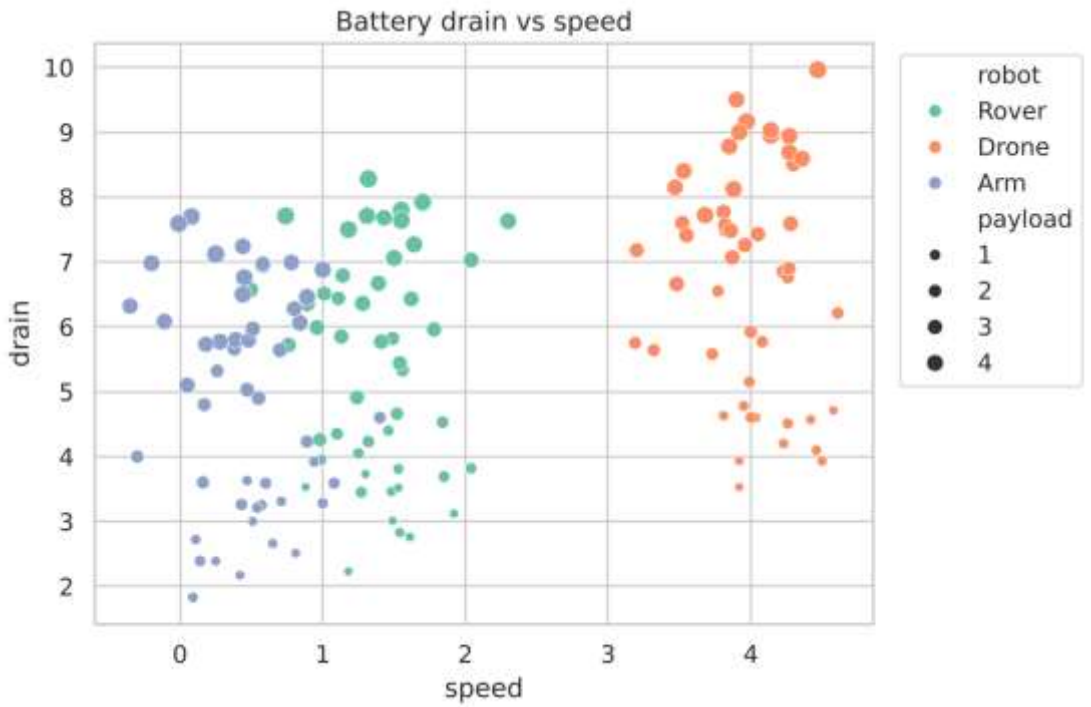
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Scatter Plot

chart.py

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="whitegrid", palette="Set2")
df = pd.read_csv("robot_sensor_data.csv")

sns.scatterplot(
  data=df, x="speed", y="drain",
  hue="robot", size="payload")
plt.legend(bbox_to_anchor=(1.02, 1))
plt.title("Battery drain vs speed")
plt.show()
```



WHAT IT SHOWS

- Each dot is one reading; position shows two numeric variables
- hue colors the dots by robot, size scales them by payload
- A rising cloud means the two variables increase together

ROBOTICS USE Check whether faster motion or heavier payloads drain the battery. All charts use robot_sensor_data.csv.

Line Plot

chart.py

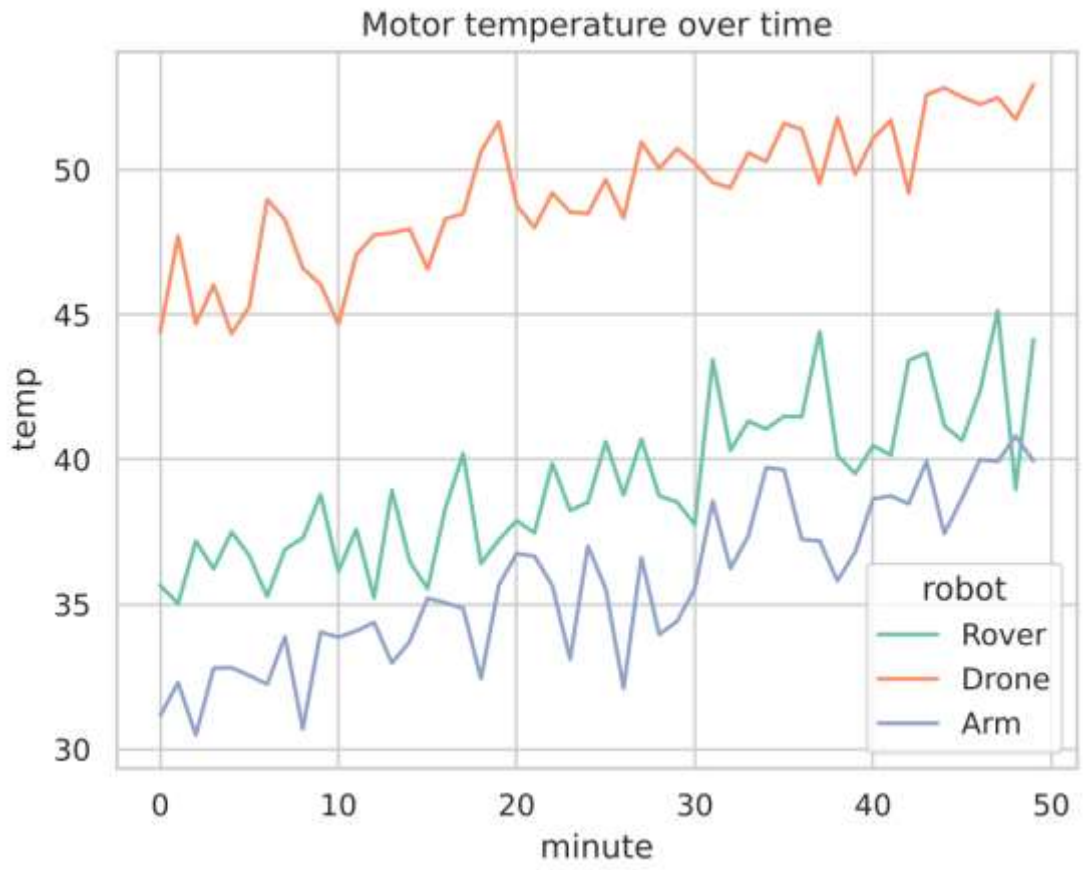
```
sns.lineplot(
  data=df, x="minute", y="temp",
  hue="robot")

plt.title("Motor temperature over time")
plt.show()
```

WHAT IT SHOWS

- Shows how a value changes along an ordered axis such as time
- Seaborn draws one line per hue category
- The shaded band is the confidence interval around the average

ROBOTICS USE Watch motor temperature climb during a mission and spot overheating early.



Histogram

chart.py

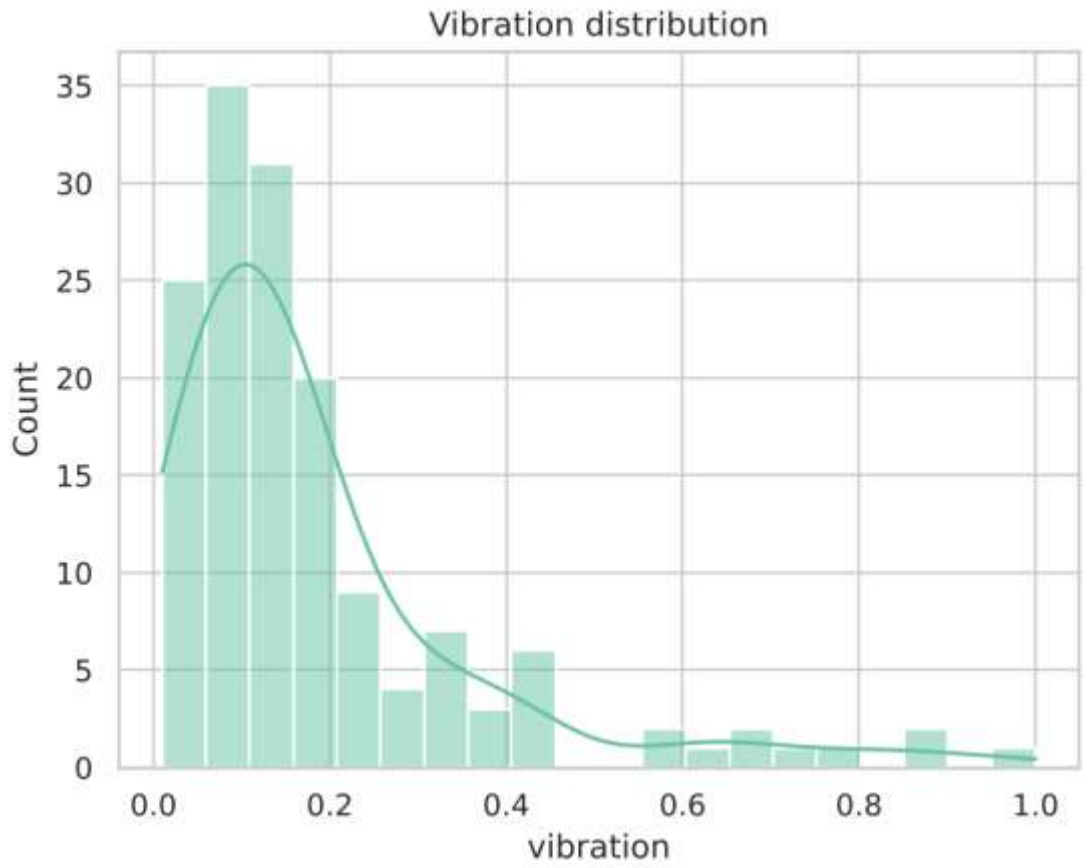
```
sns.histplot(
  data=df, x="vibration",
  bins=20, kde=True)

plt.title("Vibration distribution")
plt.show()
```

WHAT IT SHOWS

- Bars count how many readings fall in each value range (bin)
- bins controls how coarse or fine the ranges are
- kde=True adds a smooth curve estimating the distribution

ROBOTICS USE See what normal vibration looks like and spot a long tail of bumpy readings.



KDE Plot

chart.py

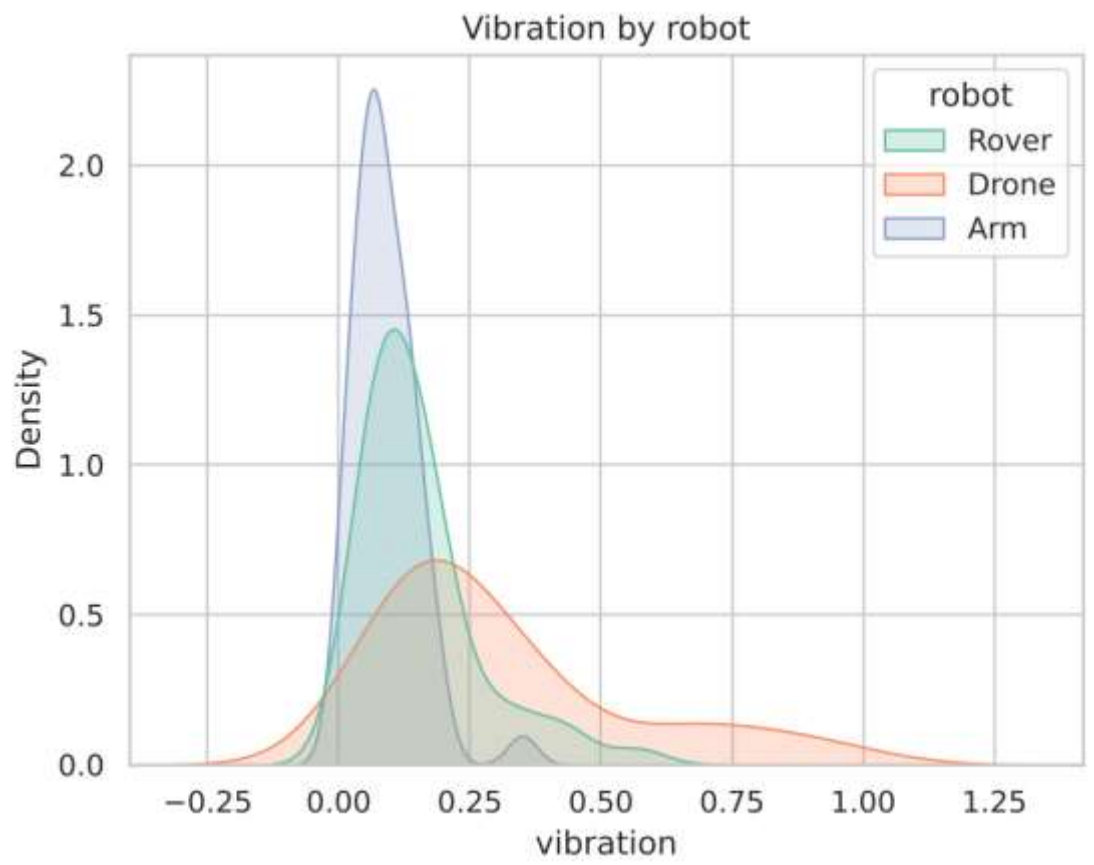
```
sns.kdeplot(
  data=df, x="vibration",
  hue="robot", fill=True)

plt.title("Vibration by robot")
plt.show()
```

WHAT IT SHOWS

- A kernel density estimate draws a smooth curve instead of bars
- hue overlays one curve per category so shapes can be compared
- Higher curve means more readings around that value

ROBOTICS USE Compare vibration profiles of different robots on one chart.



CATEGORICAL

Box Plot

chart.py

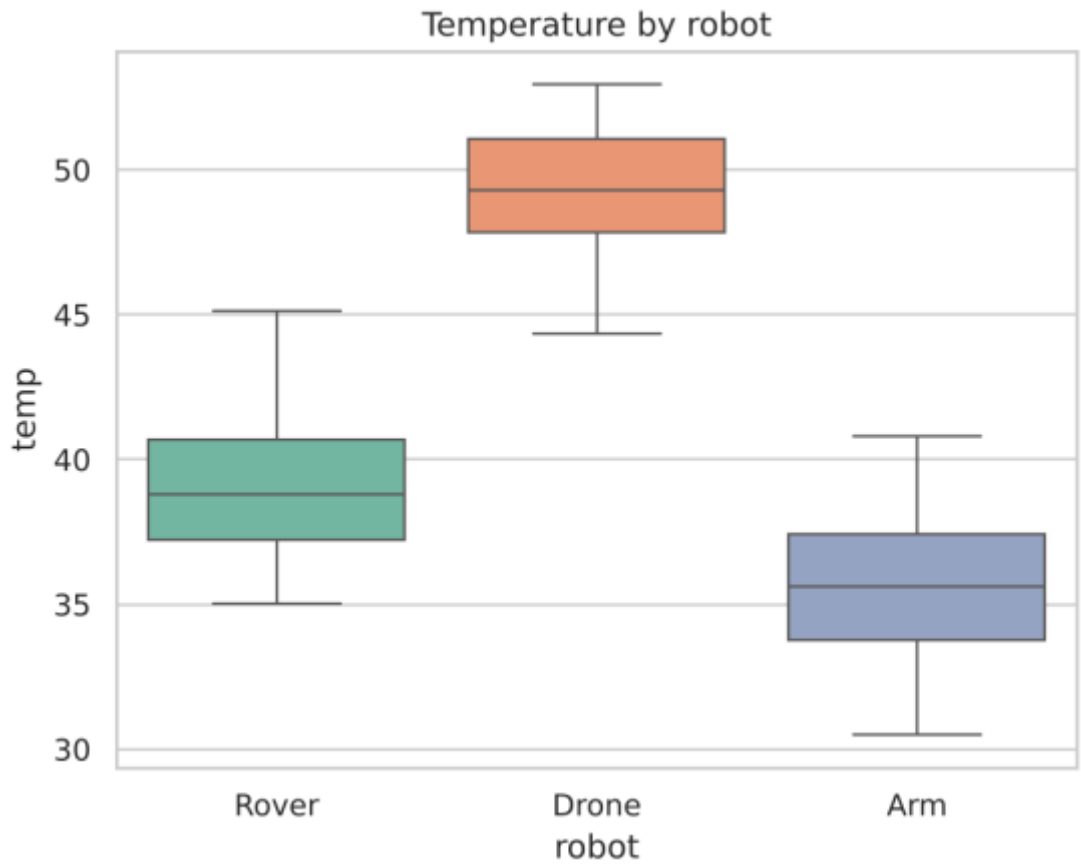
```
sns.boxplot(
  data=df, x="robot", y="temp",
  hue="robot")

plt.title("Temperature by robot")
plt.show()
```

WHAT IT SHOWS

- The box spans the middle 50% of values, with the line at the median
- Whiskers reach the typical range; dots beyond are outliers
- Quickly compares spread and centre across categories

ROBOTICS USE Find which robot runs hottest and which has unusual temperature spikes.



CATEGORICAL

Violin Plot

chart.py

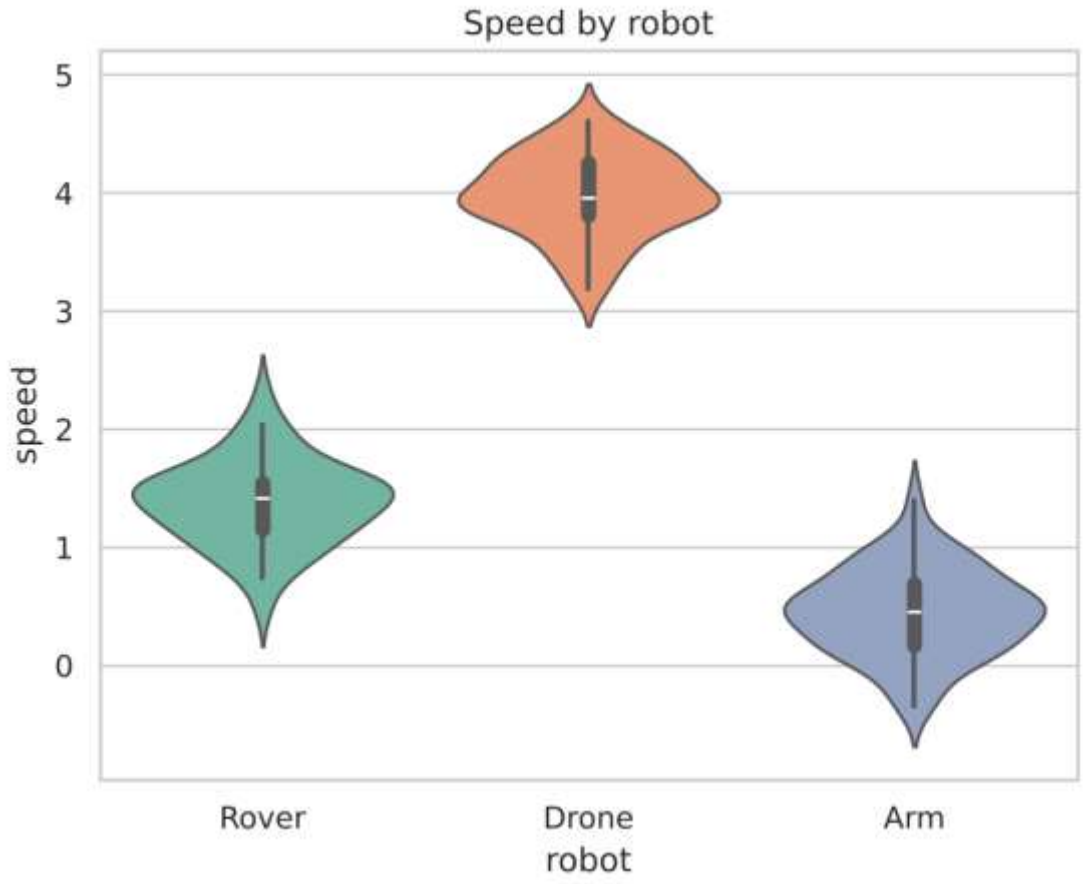
```
sns.violinplot(
  data=df, x="robot", y="speed",
  hue="robot")

plt.title("Speed by robot")
plt.show()
```

WHAT IT SHOWS

- Combines a box plot with a density curve mirrored on both sides
- Wide parts mean many readings at that value
- Reveals shapes a box plot hides, such as two peaks

ROBOTICS USE Compare how each robot's speed is distributed, not just its average.



CATEGORICAL

Bar Plot

chart.py

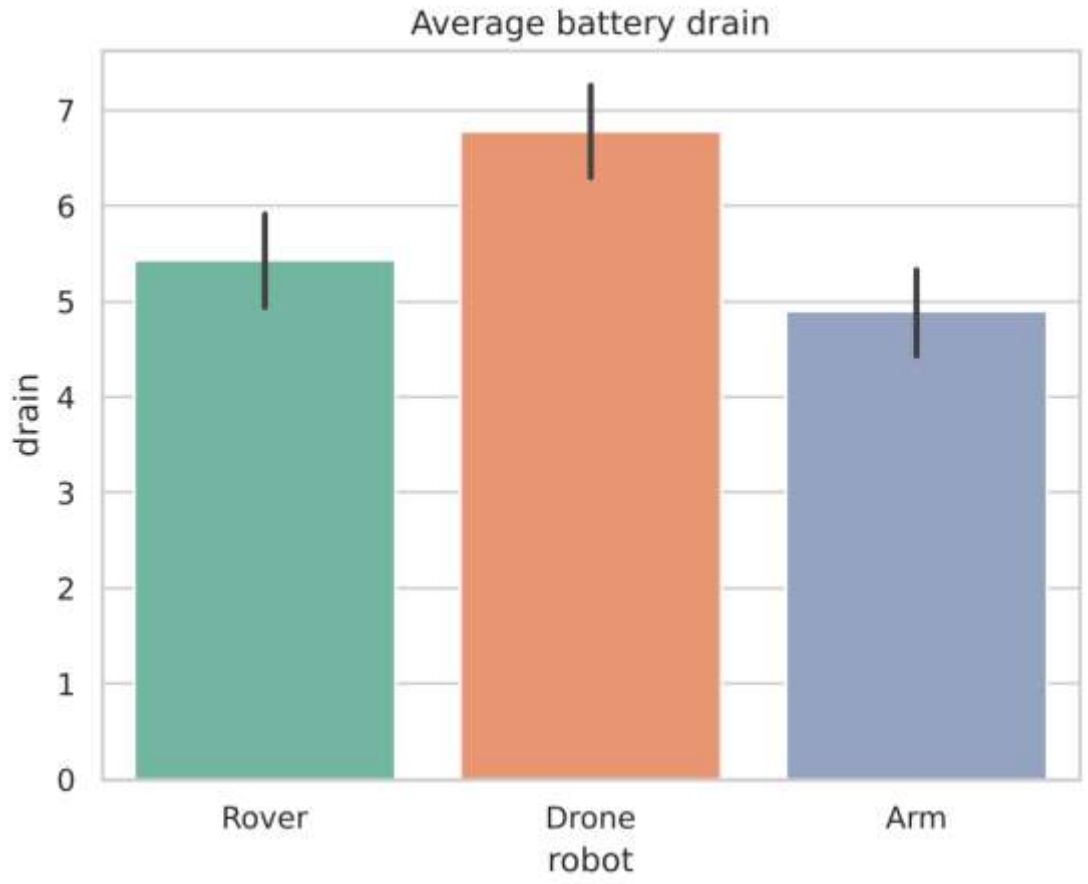
```
sns.barplot(
  data=df, x="robot", y="drain",
  hue="robot")

plt.title("Average battery drain")
plt.show()
```

WHAT IT SHOWS

- Bar height is the mean of the numeric column for each category
- The black line is the confidence interval of that mean
- Best for comparing one average across a few groups

ROBOTICS USE Rank robots by average battery drain.



CATEGORICAL

Count Plot

chart.py

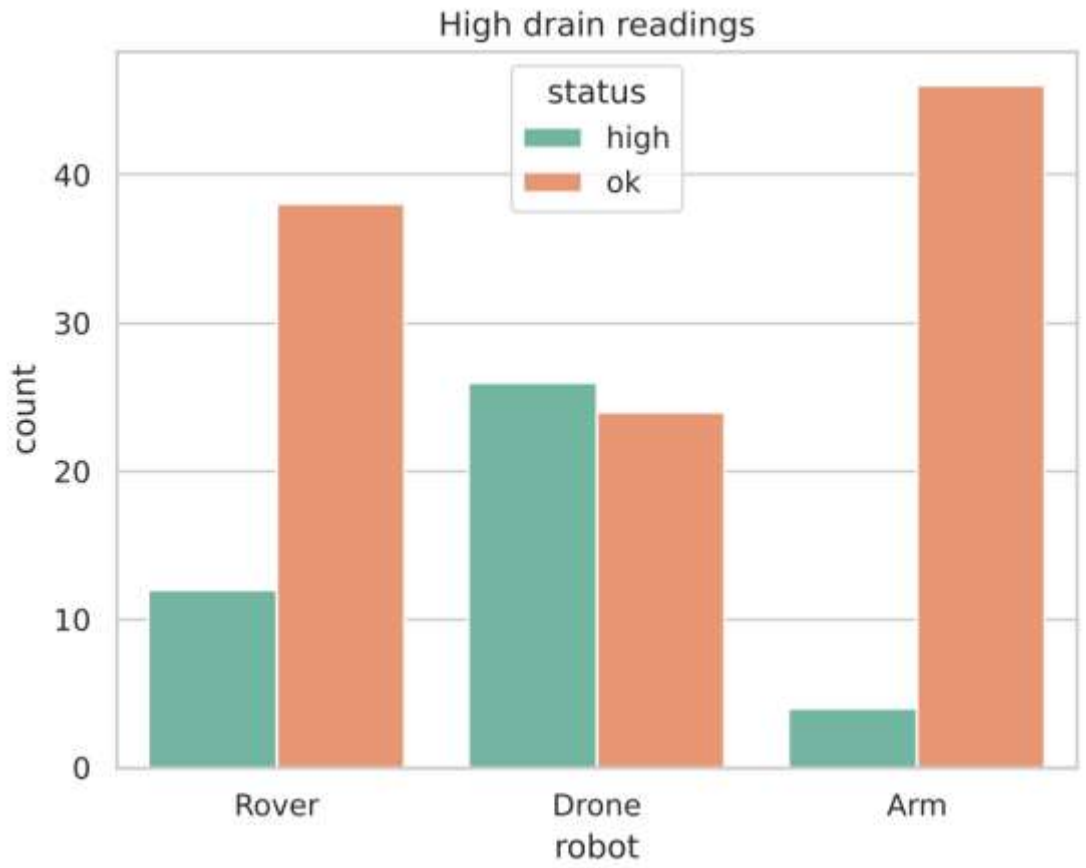
```
sns.countplot(
  data=df, x="robot", hue="status")

plt.title("High drain readings")
plt.show()
```

WHAT IT SHOWS

- Counts how many rows fall in each category, with no y column needed
- hue splits each bar by a second category
- Good for status codes, error types and class balance

ROBOTICS USE Count how often each robot reports high battery drain, a key check before training an AI model.



CATEGORICAL

Swarm Plot

chart.py

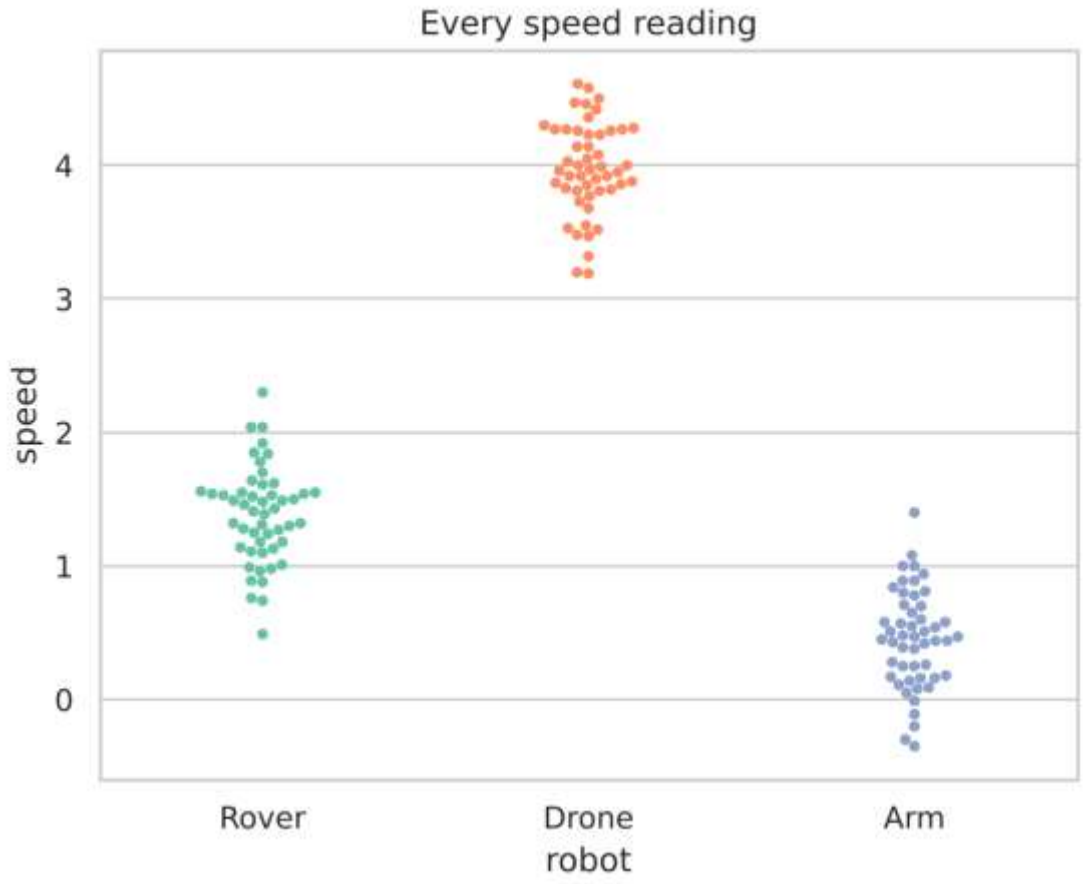
```
sns.swarmplot(
  data=df, x="robot", y="speed",
  hue="robot", size=4)

plt.title("Every speed reading")
plt.show()
```

WHAT IT SHOWS

- Shows every single reading as a dot, nudged so none overlap
- Reveals sample size and individual outliers
- Works best with small and medium datasets

ROBOTICS USE Inspect individual readings behind each robot's speed summary.



Regression Plot

chart.py

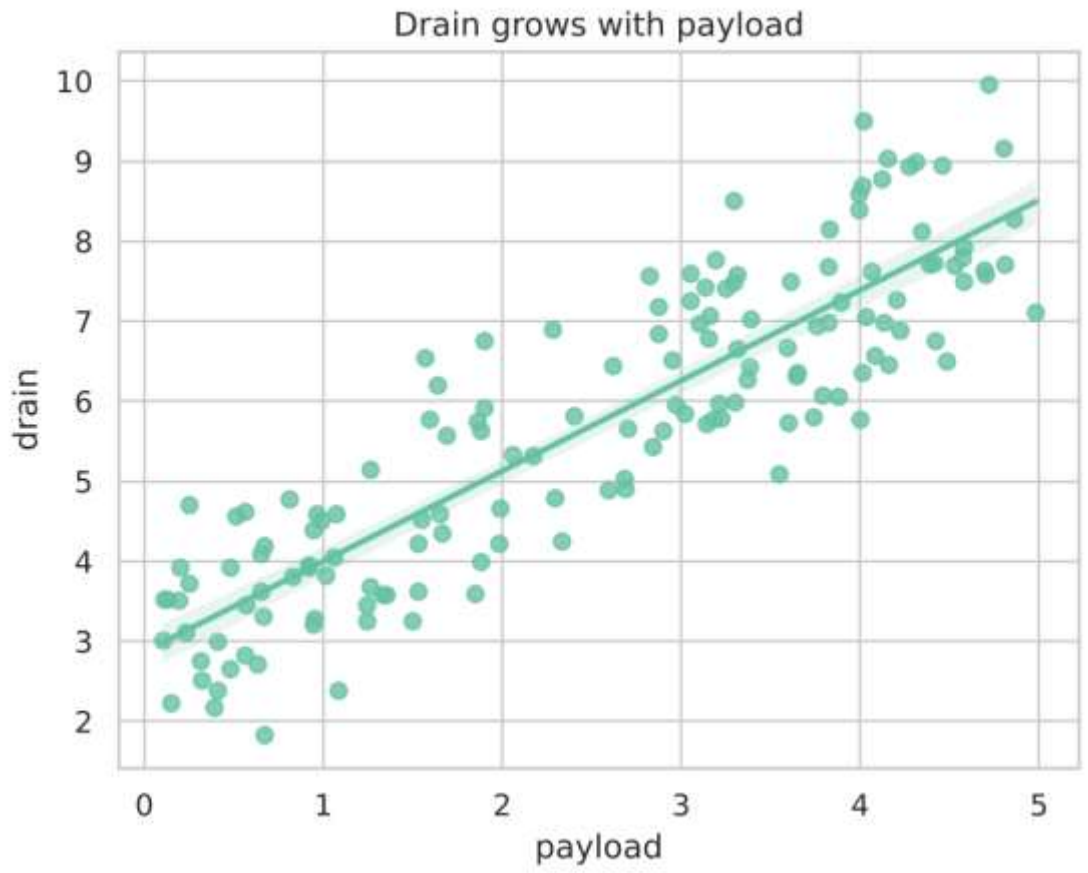
```
sns.regplot(
  data=df, x="payload", y="drain")

plt.title("Drain grows with payload")
plt.show()
```

WHAT IT SHOWS

- Draws a scatter plot plus a fitted straight regression line
- The shaded band shows uncertainty in the fit
- A steep line means payload strongly affects drain

ROBOTICS USE Estimate extra battery use for each additional kilogram of payload.



Heatmap

chart.py

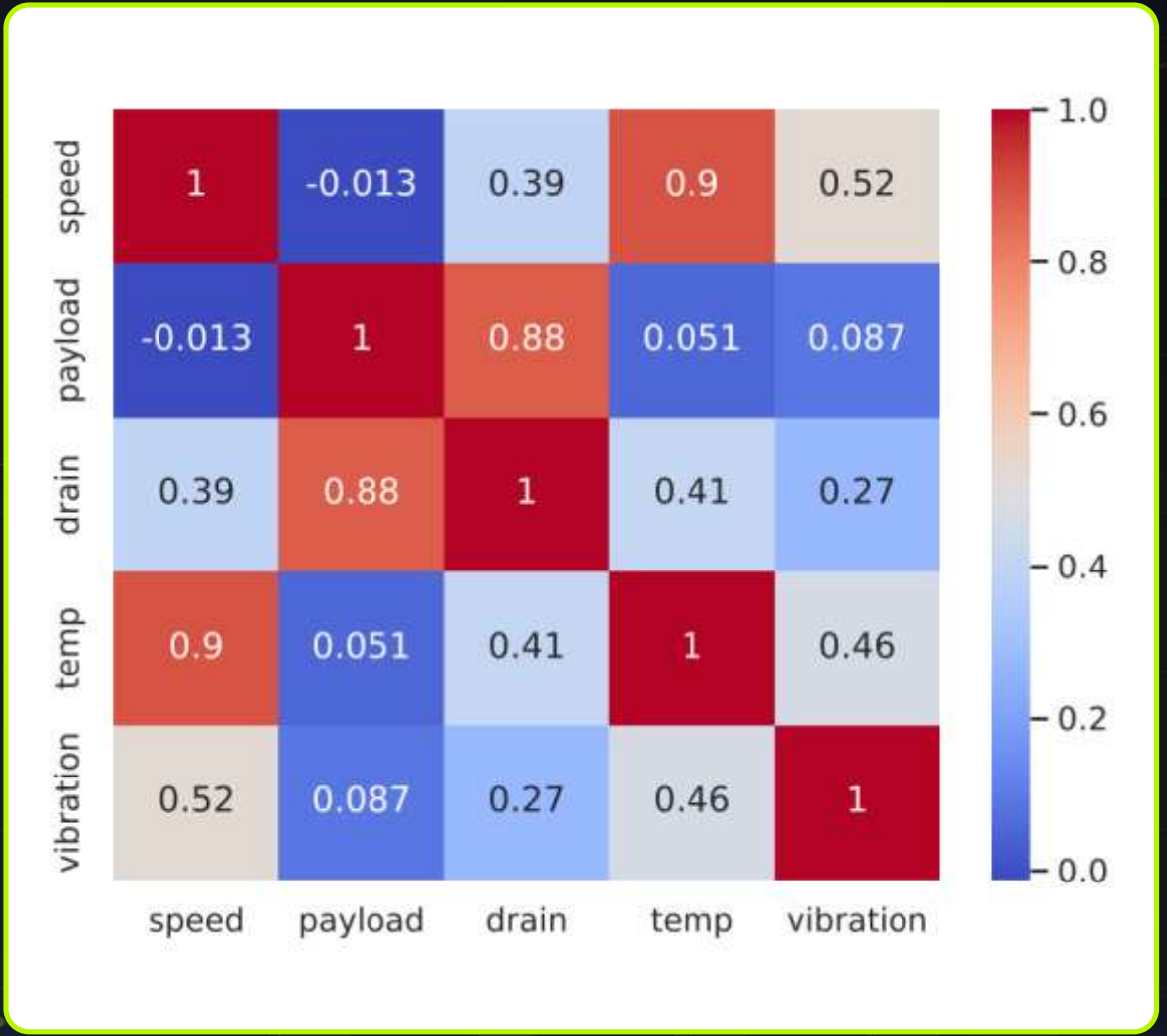
```
corr = df[["speed", "payload", "drain",
           "temp", "vibration"]].corr()

sns.heatmap(corr, annot=True,
            cmap="coolwarm")
plt.show()
```

WHAT IT SHOWS

- Colors a table of numbers: here, correlations between columns
- Values near +1 or -1 mean a strong relationship; near 0 means none
- annot=True prints each value inside its cell

ROBOTICS USE Find which sensor signals move together before choosing features for a model.



Joint Plot

chart.py

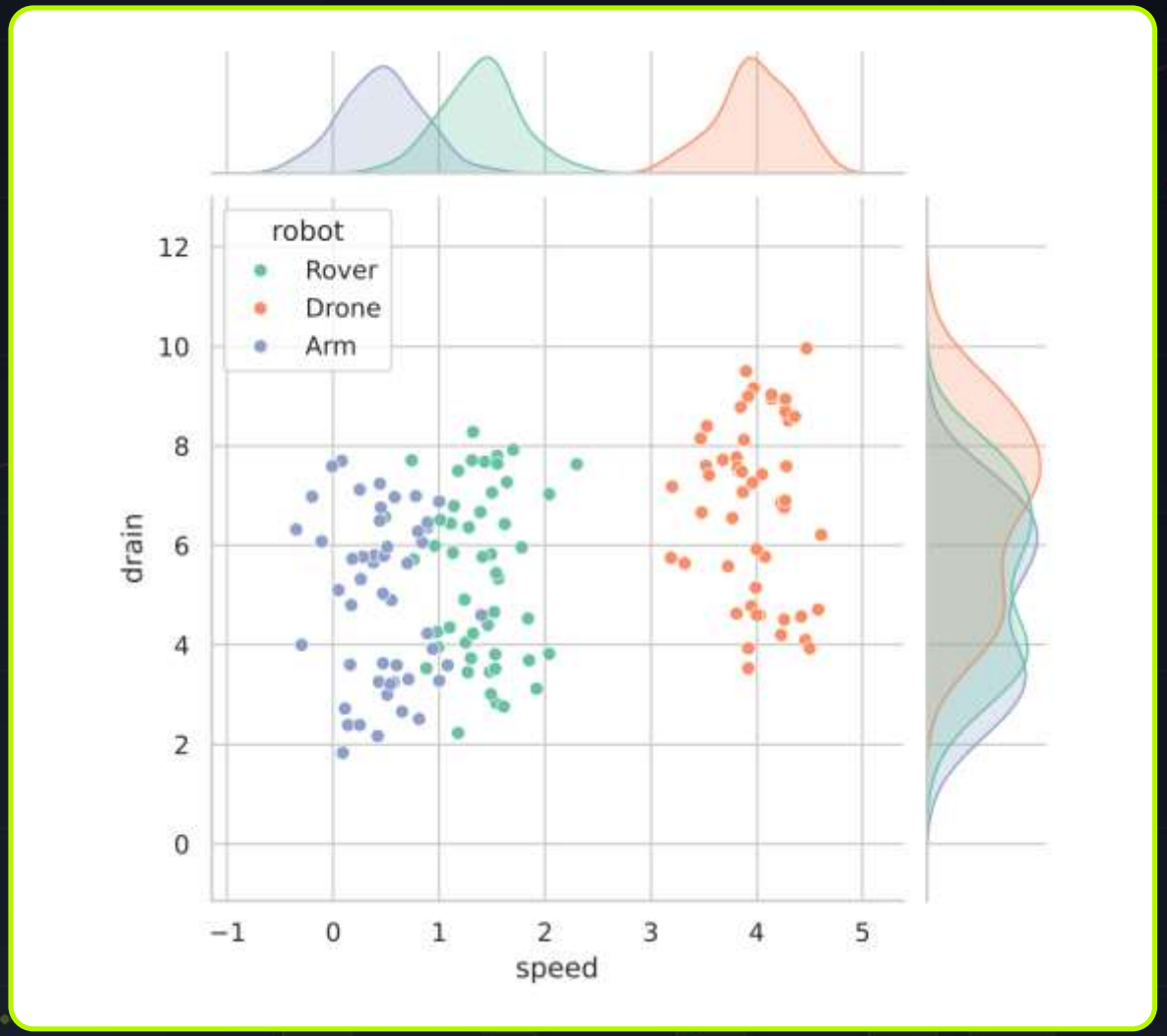
```
sns.jointplot(
  data=df, x="speed", y="drain",
  hue="robot")

plt.show()
```

WHAT IT SHOWS

- Combines a scatter plot with a histogram or density on each axis
- Shows the relationship and each variable's distribution together
- hue splits everything by category

ROBOTICS USE See how two signals relate while checking how each one is spread.



Pair Plot

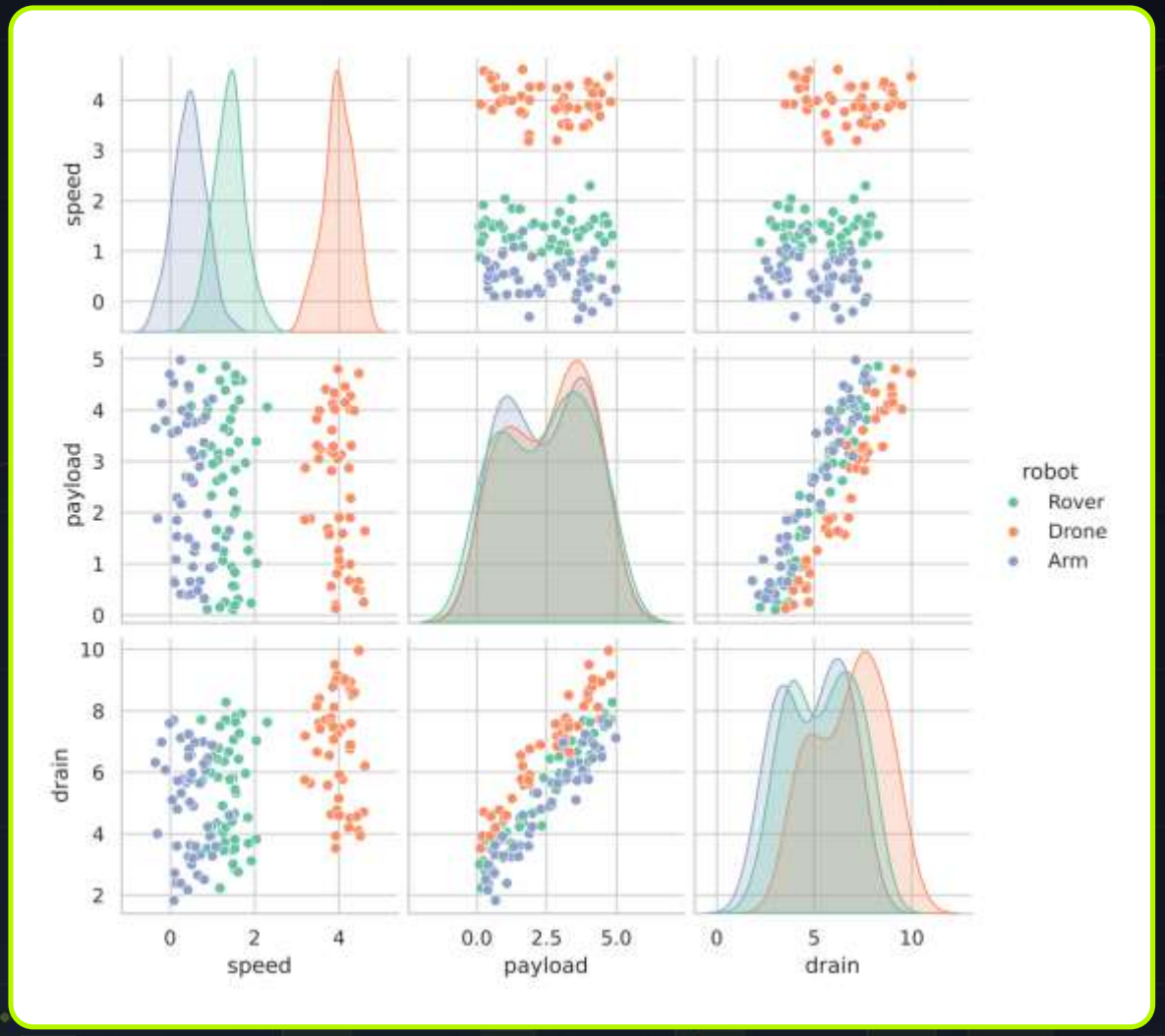
chart.py

```
sns.pairplot(
  df[["robot", "speed", "payload",
      "drain"]],
  hue="robot")
plt.show()
```

WHAT IT SHOWS

- Plots every numeric column against every other in a grid
- The diagonal shows each column's own distribution
- The fastest way to explore a new dataset

ROBOTICS USE Scan a whole sensor log at once to spot patterns and clusters.



Facet Grid (relplot)

chart.py

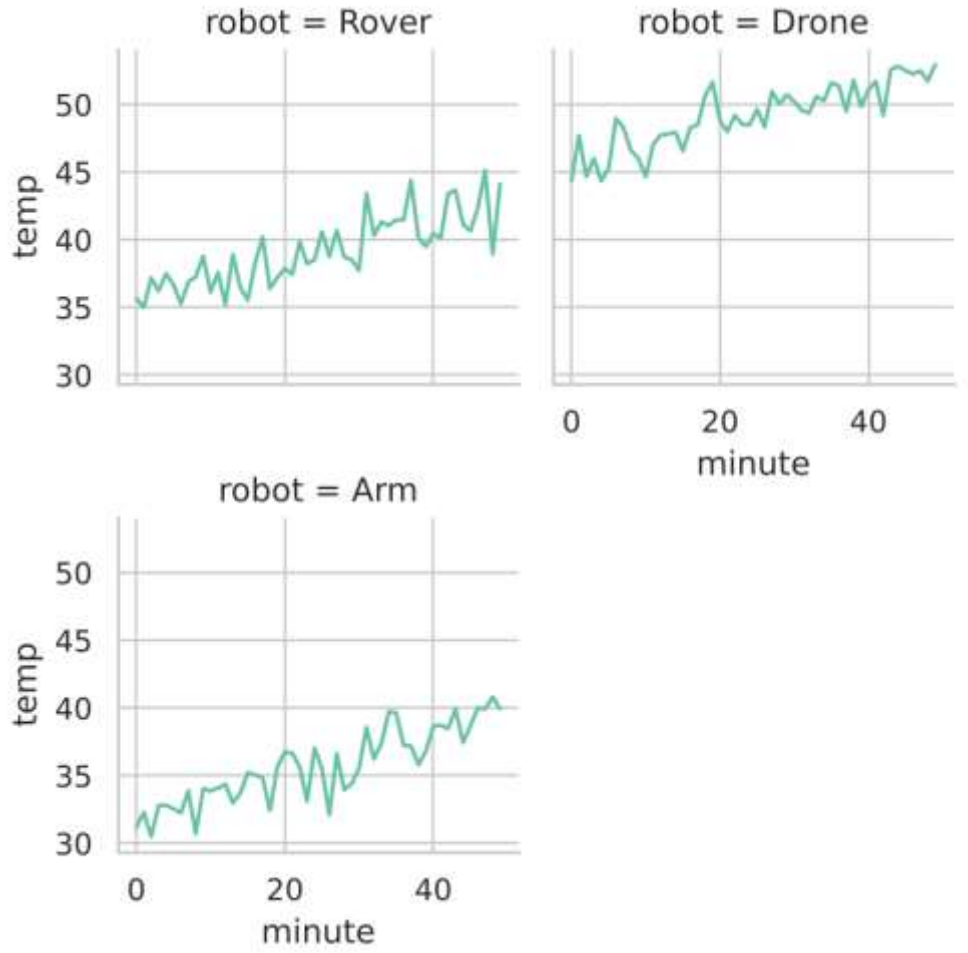
```
sns.relplot(
  data=df, x="minute", y="temp",
  col="robot", col_wrap=2,
  kind="line", height=2.6)

plt.show()
```

WHAT IT SHOWS

- Splits one chart into several panels, one per category
- col (or row) chooses the column to split by
- Every panel shares the same axes, so panels compare fairly

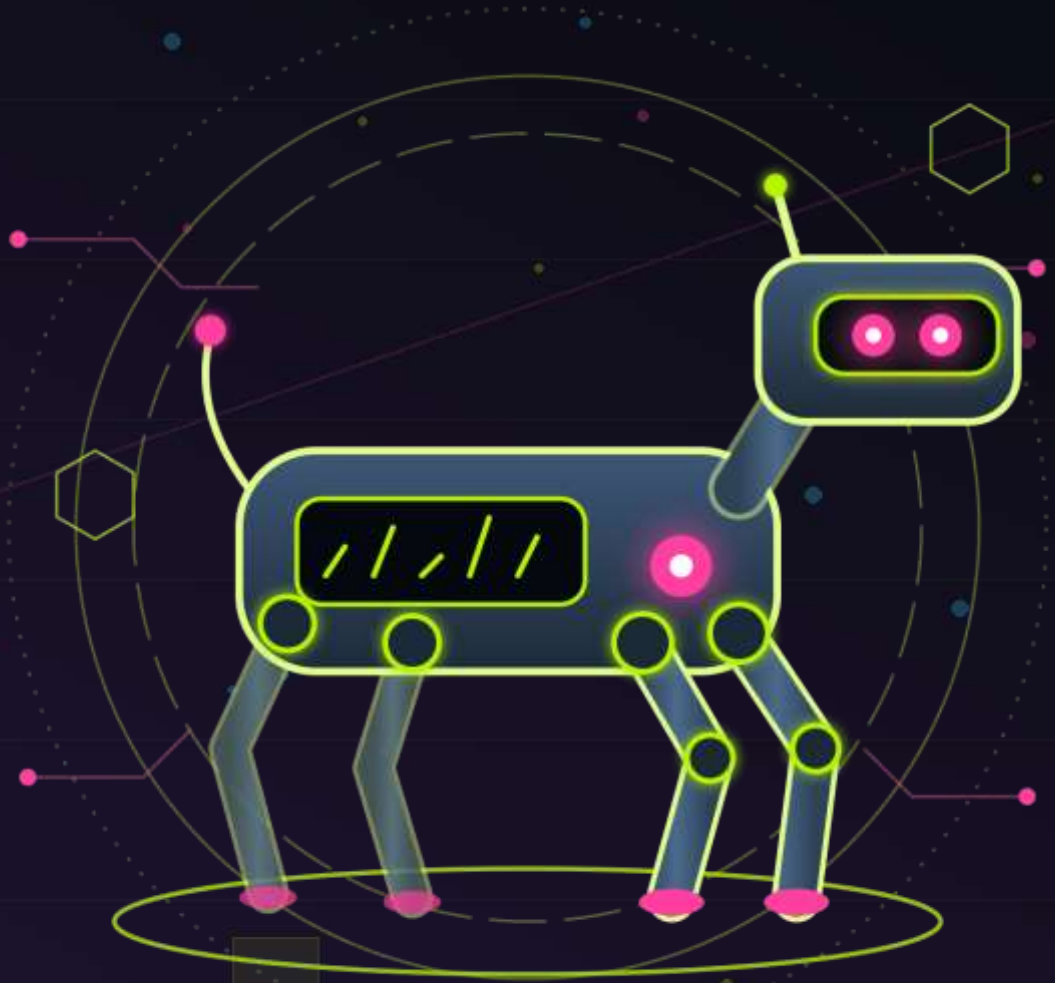
ROBOTICS USE Give each robot its own temperature panel without cluttering one chart.



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Thank You

Seaborn charts for AI & Robotics: explore your sensor data, then explain it



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